Bottle Count

Counting bottles from a phone photo



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AI for Good / Solo workshop resit

How the notebook counts bottles

1. Upload a photo

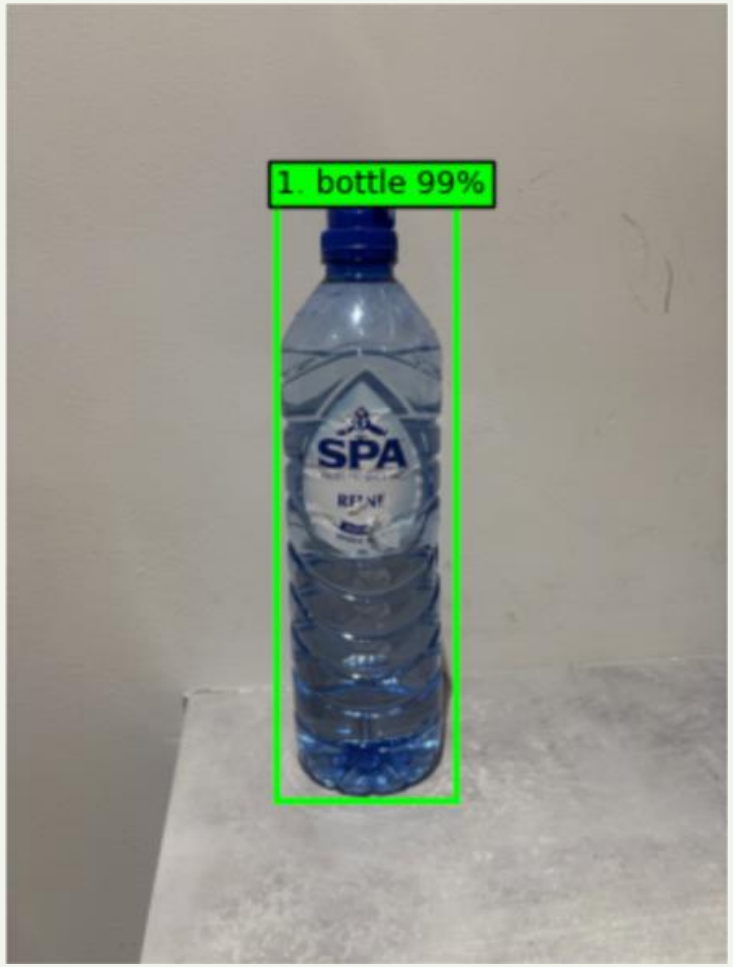
The notebook corrects the photo orientation and converts it to RGB and then to a tensor.

2. Find the bottles

Faster R-CNN with MobileNetV3 predicts a box, label and confidence score for each detection.

3. Count the boxes

Only detections labeled bottle with a score of 50% or higher are counted.



Ten photos and a manual count

What I photographed

Plastic and glass bottles, a seven-bottle lineup, overlapping bottles, different angles and dimmer light.

How I recorded the ground truth

I counted the visible bottles by hand and put the count at the end of each filename.

How I used the cartons

I included cartons next to bottles and on their own to check whether the model would count them as bottles.

09_mixed_2.jpg

2 bottles with 2 cartons

10_cartons_0.jpg

0 bottles with 2 cartons

Results at 50% confidence

10 / 10

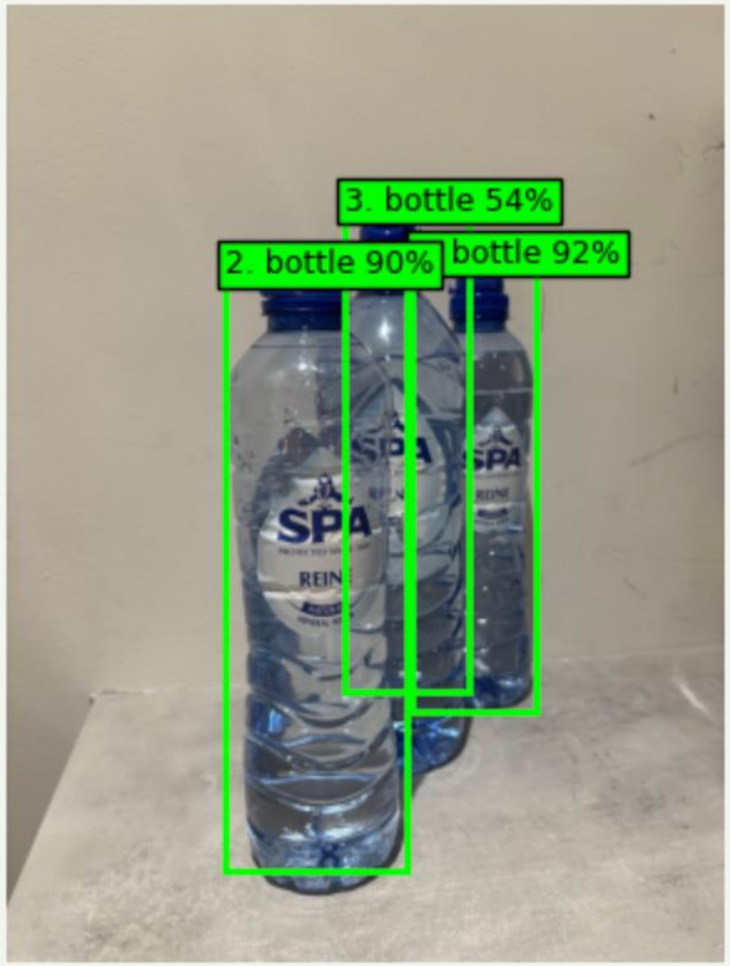
photos with an exact count

0.00

mean absolute error in bottles per photo

Example	Manual count	Model count
Seven-bottle lineup	7	7
Bottles beside cartons	2	2
Cartons only	0	0

What changes at 60%



3 bottles by hand

50% cutoff → 3 counted
60% cutoff → 2 counted

The middle bottle scored 54%, so the higher cutoff drops it. A 58% detection also drops out in the spaced photo.

At 60%, 8 of 10 counts match.

Demo and practical limits

Demo photo

The notebook drew seven boxes and returned a count of seven.

Where it could fail

Crowded shelves, strong reflections or bottles hidden behind others could lead to missed counts.

How a café worker can use it

Check the boxes against the photo and recount before updating stock if anything looks off.

